

Sofia Nikiforova

AI Researcher & ML Engineer

NLP · LLMs · Vision-Language Models · Multimodal AI · Causal AI

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I build and evaluate AI systems for real-world settings, with experience leading projects from data acquisition and exploratory analysis to model design, training, evaluation, backend implementation, and communication of results.

My work spans LLM-powered causal analysis tools, NLP for financial and biomedical information extraction, knowledge-aware image caption generation, radiology-focused multimodal AI, and sensor-based failure prediction with time series data. I bring a Ph.D. background in Computational Linguistics and a focus on reliability, interpretability, and practical usefulness.

Experience

Senior AI Researcher

Oct 2024–Present

Causify

Led AI project teams developing systems for causal inference, time series prediction, and LLM-powered data analysis. Owned end-to-end delivery from data acquisition and exploratory analysis to model design, training, evaluation, client reporting, and production-oriented implementation.

- Led development of a causal failure prediction framework for wind turbines, combining sensor time series data, domain knowledge, and predictive modeling to identify early indicators of equipment failure.
- Led development of an LLM-powered tool for generating, validating, and interacting with causal DAGs from datasets and metadata, with support for user feedback and on-demand causal analysis.
- Supervised and evaluated interns working on AI, data science, and causal inference projects.
- Contributed to research papers, technical reports, patent-related work, and internal tooling for code quality, testing, and reproducibility.

Postdoctoral Researcher — AI

Mar 2023–Oct 2024

Utrecht University

Developed multimodal image-text AI systems for radiology, with a focus on integrating visual and linguistic representations for clinical decision-support applications.

- Designed and implemented models combining medical imaging with structured and unstructured textual data.
- Emphasized factual accuracy, interpretability, and robustness in user-facing AI systems for high-accuracy domains.
- Collaborated with an interdisciplinary team across AI, medicine, and data science.

Senior Data Scientist — NLP

Nov 2016–Jun 2021

Particle.One

Led the development of NLP-driven classification, knowledge extraction, and predictive modeling systems for the financial domain.

- Owned end-to-end NLP model development, from problem formulation and data acquisition to model design, evaluation, deployment, and monitoring.
- Built systems for text classification, information extraction, and predictive analytics.
- Translated business and data problems into applied machine learning solutions.

Teaching Assistant

2021

Utrecht University

Course: Machine Learning for Human Vision and Language (MSc Artificial Intelligence).

Developed and graded assignments and supervised practical lab sessions covering state-of-the-art machine learning techniques for vision and language tasks.

NLP Developer

2017–2018

Semantic Hub

Developed NLP tools for information extraction in the life sciences/biomedical domain. Implemented pipelines for entity recognition, relation extraction, and structured data generation from heterogeneous unstructured text, including technical documents and user-generated content.

Education

Ph.D., Computational Linguistics

2018–Feb 2023

Utrecht University, Netherlands

Dissertation: *Image Captioning with External Knowledge*

Ph.D. research focused on knowledge-aware image captioning: vision-language models for grounded generation of image descriptions enriched with relevant external world knowledge. Developed methods for identifying image-relevant entities, retrieving contextual

facts from external resources, and integrating this knowledge into caption generation, with a focus on factual accuracy. The work included three captioning models across geographic and news-image domains and two newly compiled datasets.

M.A., Computational Linguistics

Higher School of Economics, Moscow, Russia

2016–2018

B.A., Fundamental and Applied Linguistics

Lomonosov Moscow State University (MSU), Russia

2012–2016

Selected Publications

- **Causify DataMap: Automatic Causal Probabilistic Reasoning.** *Internal Causify report, 2026.*
LLM-powered system for generating, validating, and interacting with causal DAGs from datasets and metadata, with support for user feedback and on-demand causal analysis.
- **DMCD: Semantic-Statistical Framework for Causal Discovery.** *arXiv preprint, 2026.* [arXiv:2602.20333](#)
A two-phase causal discovery framework that integrates LLM-based semantic drafting from variable metadata with statistical validation on observational data.
- **Runnable Directories: The Solution to the Monorepo vs. Multi-repo Debate.** *arXiv preprint, 2025.* [arXiv:2512.03815](#)
Software-architecture pattern that reconciles trade-offs between monorepo and multi-repo organisation of code.
- **Causify Sentinel: A Causal Failure Prediction Framework.** *Internal Causify report, 2025.*
Causal framework for predicting failures in sensor time-series systems by combining domain knowledge, predictive modeling, and causal reasoning.
- **Generating Image Captions with External Encyclopedic Knowledge.** *arXiv preprint, 2022.* [arXiv:2210.04806](#)
Image captioning approach that integrates encyclopedic knowledge into generation, with emphasis on factual accuracy.
- **Geo-Aware Image Caption Generation.** *Proceedings of COLING 2020.* [ACL Anthology](#)
Image captioning with geographic context grounding for generating location-aware photo descriptions.

Patents

- **Systems and Methods for Causal Failure Prediction in Time Series** — U.S. Patent Application No. 63/934,790

Selected Talks

- **Imperfect LLMs in High-Accuracy Fields** — NLG Summit in the Lowlands, Utrecht University (2023)
- **On factual accuracy evaluation in knowledge-aware text generation** — NLG Summit in the Lowlands, Tilburg University (2022)
- **Geo-Aware Image Caption Generation** — COLING 2020, Barcelona (Online)
- **Geographic Context for Image Captioning** — GeCKo Symposium, Barcelona (Online), 2020

Skills & Technologies

Skills

Machine Learning · NLP · LLMs · Generative AI · Information Extraction · Model Evaluation · Multimodal AI · Vision-Language Models · Causal Inference · Time Series Modeling · Technical Leadership

Technologies

Python · PyTorch · pandas · NumPy · scikit-learn · spaCy · Git · GitHub · LaTeX · Jupyter · Linux

Languages

Russian (native) · English (fluent) · Dutch (upper-intermediate)

Location

Amersfoort, Netherlands